

CompositionProof-2 - Epoch 1 Record

Status: EPOCH 1 COLLECTED - NO FINAL VERDICT. Data collection only. A cross-witness correlation verdict requires at least $N_{MIN} = 20$ distinct calibration epochs (target 30). This is 1 of ≥ 20 .

Authorization: Derek Hone, "GO COMPOSITIONPROOF-2 - COLLECT EPOCH 1 ONLY", 2026-09-08. One epoch battery submitted, then stop.

Pre-flight (all gates passed before submission)

Gate	Result
IBM Quantum access	Restored - channel <code>ibm_quantum_platform</code> , token from secrets store, verified live (open plan)
Frozen manifest / lock	INTACT - composite <code>ebe0919b997341569670c-cd27b2a8ff754599d242489a2b06ff4db20c8872f-d6</code> , 11/11 files verified
Calibration independence	PASS - <code>ibm_marrakesh</code> calibration 2026-09-08T16:59:31Z; no prior CP-2 epoch existed, so timestamp is distinct
Budget (pre)	363 s remaining of 600 s (28-day rolling)

Device choice: `ibm_marrakesh` (primary; same named device as CompositionProof-1 - the longitudinal series must stay on one device). Queue was empty (pending_jobs=0).

Epoch 1 result

Field	Value
Epoch ID	1
Job ID	dag4fnfi3e6s738ltnpg
Backend	ibm_marrakesh (Heron r2, 156 qubits)
Calibration timestamp	2026-09-08T16:59:31+00:00
Circuits x shots	8 x 4000 = 32,000 shots
Submitted (UTC)	2026-09-08T17:37:33Z
Job status	DONE
Actual QPU charge	11 s (job metrics qpu_charge_time_seconds)
Budget after	352 s remaining of 600 s

Witness values (locked estimators from composition2_sim.py)

- **CHSH S = 2.7205** - $|E(a_0b_0) - E(a_0b_1) + E(a_1b_0) + E(a_1b_1)|$ (classical ≤ 2 , Tsirelson ~ 2.828 , single-epoch threshold 2.2). Above classical bound and above threshold.
- **Mermin-3 M = 3.7710** - $|E(XYY) + E(YXY) + E(YYX) - E(XXX)|$ (classical ≤ 2 , quantum 4, single-epoch threshold 2.6). Above classical bound and above threshold.

Per-setting expectations:

- CHSH E: $a_0b_0 = 0.6770$, $a_0b_1 = -0.6745$, $a_1b_0 = 0.6630$, $a_1b_1 = 0.7060$

- Mermin E: $XYY = -0.9495$, $YXY = -0.9415$, $YYX = -0.9355$, $XXX = 0.9445$

Calibration snapshot (medians across 156 qubits): $T1 = 168.87$ us, $T2 = 66.95$ us, readout error = 0.011475.

Replication note (context, not a verdict): CompositionProof-1 on the same device measured $S = 2.7765$, $M = 3.7365$. Epoch 1's $S = 2.7205$, $M = 3.7710$ are consistent - both witnesses remain strongly nonclassical in this calibration state.

What this is NOT

- This is **NOT** a PASS or FAIL for CompositionProof-2. The experiment tests whether CHSH and Mermin-3 epoch-to-epoch fluctuations are correlated across many distinct calibration epochs. A single epoch has no fluctuation series and no correlation estimate.
- No claim of correlation, independence, or common cause is made or implied from one epoch.
- Honesty bounds (from the frozen lock) hold: device-dependent witnesses on co-located qubits; locality and detection loopholes OPEN; not device-independent; not loophole-free.

Next step (requires a genuinely new calibration epoch)

The independence guard in `submit_epoch.py` will REFUSE a second battery that shares this calibration timestamp. The next epoch may be submitted only after `ibm_marrakesh` recalibrates (a new `last_update_date`), typically ~1 per calibration cycle. Collection proceeds ~1 epoch per cycle until ≥ 20 distinct epochs, then `verifier2.py` emits the frozen cross-witness verdict. No further submission is authorized beyond Epoch 1 under the current GO.